

ORIGINAL ARTICLE

Federated Deep Learning for Collision Avoidance in IoV With Digital Twin Integration

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ABSTRACT

The Internet of Vehicles (IoV) is revolutionising transportation by connecting vehicles, infrastructure and devices, enabling more intelligent and safer mobility. One key challenge is ensuring efficient and secure communication among vehicles with varying capabilities, including different sizes, speeds and sensor configurations. This research introduces a Federated Learning-Driven Deep Learning (FLDL) approach to intelligent collision avoidance, designed to address the heterogeneity of vehicles in the IoV ecosystem. The system integrates real-time data from vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications, while considering factors like vehicle type, road conditions, driver behaviour and Digital Twins. Our approach leverages multiple Federated Learning strategies, which enhance privacy protection, reduce communication overhead and enable real-time decision-making without the need for centralised data storage. Experimental results show that the GNN + FedGC model achieves the highest performance with an accuracy of 98.8%, outperforming other models such as MLP with FedLU (98.5%), DRL with FedPPO (98.3%) and LSTM with FedSGD (97.65%). The integration of Digital Twins further enhances model accuracy by simulating real-time vehicle behaviour and environmental conditions. This FL-based system not only improves collision prediction but also enhances safety, reduces accident rates and supports scalable decision-making in smart city transportation systems.

1 | Introduction

The rapid evolution of the Internet of Things (IoT) has profoundly impacted various industries, paving the way for the advent of new technologies that facilitate smarter and more efficient systems. The IoV is one of the most promising applications based on the IoT, enabling automobiles to communicate not only with each other

but also with the surrounding infrastructure, as well as with other people and objects within the transportation ecosystem (Herich and Vaščák 2024; De Vincenzi et al. 2024). The IoV is designed to establish a more efficient system of communication between vehicles, infrastructure, and surroundings, thereby generating a better driving experience, enhancing road safety and reducing traffic congestion. IoV has the potential to create a smarter transportation

network that responds to the dynamism and complexity of real-time data exchange. The IoV system integrates multiple communication schemes, including V2V, V2I, Vehicle-to-Everything (V2X) and Vehicle-to-Grid (V2G), which continuously transmit essential information such as traffic and road conditions, as well as vehicle patterns and data from digital twins, to enhance real-time situational awareness. To illustrate, V2V communication enables vehicles to alert each other in the event of an imminent collision, whereas V2I communication facilitates the integration of traffic and road situations with vehicles through traffic signals and other means. Consequently, IoV has the potential to greatly contribute to road safety, increase traffic efficiency and reduce road accidents (Ji et al. 2022; Na et al. 2024). In Table 1, a list of abbreviations is provided with definitions of abbreviations that are frequently used in the course of writing the paper.

There are numerous challenges to adopting IoV, despite its tremendous potential. One of them is vehicular heterogeneity, which means the different types, sizes, speeds, capabilities and sensor configurations of vehicles. Differences in the quality and consistency of the data transmitted between vehicles and

infrastructure can lead to poor communication efficiency and even safety risks. There are also human factors, such as driver behaviour, fatigue and distractions, which complicate the real-time prediction of accidents and their prevention (Muzahid et al. 2022; Rajagopal et al. 2023). FL is an emerging excellence in tackling these problems through distributed scenarios in multiple vehicles. FL enables each vehicle to train its local model using its data and report back (not raw data) model updates to a central server, which combines the updates to enhance the global model. The process not only guarantees the privacy of sensitive data but also reduces the bandwidth required to transmit large amounts of data through the network, making it more efficient for real-time applications in IoV (Wu et al. 2022; Su et al. 2022; Adnan, Syed, et al. 2025). Figure 1 presents a conceptual diagram of the vehicular ad hoc network (VANET) architecture as a part of the IoV ecosystem. It displays some of the communication models utilised in IoV, including V2V, V2I, Vehicle-to-Pedestrian (V2P), Vehicle-to-Sensor (V2S) and Vehicle-to-Roadside Unit (V2R). These communication paradigms enable the free flow of data among vehicles, infrastructure, sensors and pedestrians, ensuring road safety, easing traffic flow and facilitating real-time decision-making (Haider et al. 2025; Khan et al. 2024).

TABLE 1 | List of abbreviations.

Abbreviation	Definition
IoV	Internet of vehicles
FL	Federated learning
CNN	Convolutional neural networks
LSTM	Long short-term memory
DRL	Deep reinforcement learning
MLP	Multilayer perceptron
GNN	Graph neural networks
V2V	Vehicle-to-vehicle
V2I	Vehicle-to-infrastructure
V2X	Vehicle-to-everything
V2G	Vehicle-to-grid
V2P	Vehicle-to-pedestrian
SVM	Support vector machine
FedAvg	Federated averaging
FedSGD	Federated stochastic gradient descent
FedPPO	Federated proximal policy optimisation
FedLU	Federated learning with local update
FedGC	Federated graph convolution
RSU	Roadside unit
ADAS	Advanced driver assistance systems
HB	Human behaviour
RSS	Road surface status
IoT	Internet of things
ML	Machine learning

We develop a novel collision avoidance system IoV based on the duality of IoV and the idea of FL, DL models and digital twins to constantly combat the aspects of vehicle heterogeneity, ineffective communications and human factors. We concentrate on the combination of five state-of-the-art deep neural network models, that is, CNN, LSTM networks, DRL, MLP and GNN. All models are tailored to address a specific collision forecast feature, taking into account the variance of IoV environments. FL techniques, including FedAvg, FedSGD and FedPPO, are employed to enable the system to continually improve as it incorporates model updates across various vehicles and digital twins, ensuring continuous learning without compromising data privacy. The motivation behind this work is to apply FL to collision avoidance systems (CASs) in the IoV, aiming to enhance privacy, scalability and real-time responsiveness. FL enables vehicles to train models collaboratively without sharing raw data, preserving user privacy and reducing the risk of data breaches. By transmitting only model updates, it minimises communication overhead and latency, making the system more efficient for real-time decision-making in dynamic traffic environments. Moreover, FL supports scalability as IoV networks grow, allowing decentralised training across thousands of vehicles. It also adapts to vehicular heterogeneity by enabling local model updates tailored to different vehicle types and sensor capabilities. This ensures accurate and personalised collision predictions across a wide range of scenarios, making FL a robust and flexible solution for next-generation intelligent transportation systems.

1.1 | Contributions

Our proposed system has several key contributions:

- Integrating FL with DL models (CNN, LSTM, DRL, MLP and GNN) and digital twins to develop a scalable, efficient and privacy-preserving collision avoidance framework for

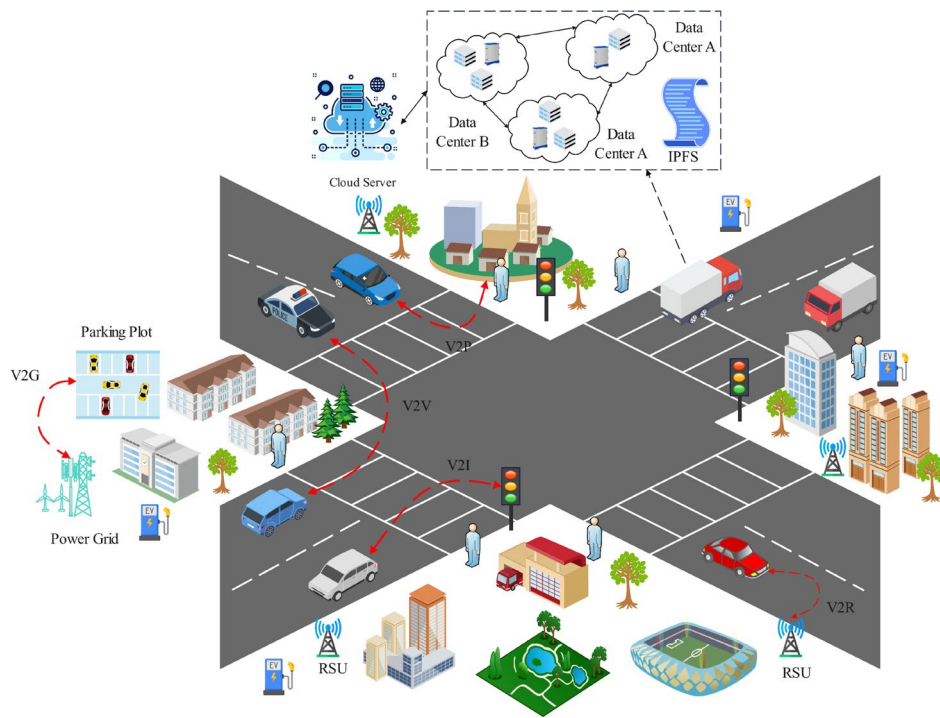


FIGURE 1 | Architectural overview of the IoV ecosystem.

the IoV, while addressing vehicular heterogeneity in terms of vehicle types, sensor configurations and capabilities.

- Addressing the diverse characteristics of vehicles, such as size, speed, manoeuvrability and sensor diversity, along with human behaviour factors (HBF) like driver distraction and fatigue, which have not been adequately considered in prior IoV CASs.
- Utilising real-time data from V2V and V2I communication to adapt dynamically to changing environments and traffic conditions, enabling timely predictions and proactive collision prevention, while factoring in the heterogeneity of vehicles and road conditions.
- Applying various FL techniques, including FedAvg, FedSGD, FedLU, FedGC and FedPPO, to improve model accuracy while maintaining privacy, reducing communication overhead, and effectively handling heterogeneous data from different vehicles.

The remainder of this paper is organised as follows: Section 2 reviews the state-of-the-art CASs and FL methods. In Section 3, the methodology is presented, discussing the machine learning (ML) models and techniques used in FL. The experimental setup is explained in Section 4, consisting of datasets, sensors and evaluation metrics. Section 5 presents the results and a comparison of the proposed system's performance with that of others, and discusses potential future directions for this research study. Finally, Section 6 concludes the study.

2 | Related Work

IoV has played a crucial role in modern transportation infrastructure, giving improved road safety, efficiency and driving experiences. The technologies used in IoV include sophisticated

communication models, like V2V, V2I and V2X. The essence of these models extends beyond communication among vehicles, as they involve exchanging urgent information regarding road conditions, traffic conditions and vehicle conditions, which are crucial for improving situational awareness and preventing accidents. IoVs present several critical challenges associated with the heterogeneity of vehicles, in terms of size, speed, sensor deployments and communication schemes (Adnan, Umer, et al. 2025; Noor-A-Rahim et al. 2022). This heterogeneity issue is particularly significant in CASs, where various vehicles cooperate despite differences in capabilities and sensor technology. As an example, a big vehicle like a truck might be communicating in a different period or type of sensor compared to smaller vehicles, such as motorcycles or passenger vehicles. That is why it is essential to consider adaptive CASs that can handle such traffic variability in the real world. The use of ML, specifically the DL, has proven very promising in this domain due to its ability to utilise large data sets sourced in various ways and make predictions that have led to real-time collision avoidance (Li, Wang, et al. 2023; Ali et al. 2024; Chhabra et al. 2023).

The authors in the work (Elahi et al. 2025; Xu et al. 2023; Onesimu et al. 2021) illustrated how FedAvg was used in a CAS, where vehicles trained locally on their own data and sent their model updates to the server, which were then combined. This solution enabled the system to preserve privacy while enhancing the accuracy of collision prediction. Likewise, study (Elahi et al. 2025) implemented FedSGD (Federated Stochastic Gradient Descent) for IoV collision prediction, updating the global model in a decentralised manner without compromising data security. The results indicated that FL methods produce high accuracy in calculating real-time collision avoidance predictions. One of the most significant issues in IoV systems is how to incorporate HB parameters into collision avoidance algorithms. Driver fatigue and distractions have been reportedly of great significance in

road accidents (Sun et al. 2023; Bakheet and Al-Hamadi 2022), and included in models in order to enhance the accuracy of predictions based on the influence they exert on vehicle behaviour. The authors proposed a DL model that helps to study the behaviour of the driver, such as eye movements and yawning, and alerts the driver, which minimises the chances of a collision as a result of fatigue. Other researchers like (Onesimu et al. 2021) used a comparable CNN to identify the distraction in the driver by the observation of facial expression and eye movement on a real-time basis.

The authors explored the use of Graph Neural Networks (GNNs) for anomaly detection and event recognition in dynamic environments. In (Ullah et al. 2023), GNNs were applied for video anomaly detection, improving recognition of spatiotemporal patterns in real time. Similarly, (Ullah et al. 2025) introduced Split Federated Learning (SFL) for UAV-based video event recognition, optimising computational tasks between UAVs and central servers to enhance real-time processing and communication efficiency. These approaches demonstrate how GNNs and Federated Learning frameworks are leveraged for complex, decentralised systems, enhancing prediction accuracy and reducing communication overhead. In studies (Chen et al. 2024; Park et al. 2025), the authors proposed a system that estimates the future position of a vehicle to prevent accidental collisions resulting from lane switching. The system contemplated recording the vehicle position and the possible conflicts using a real-time object detection project overlapped with DL models to predict the position of the vehicle. The results demonstrated that the system was able to accurately predict vehicle positions before any accidents could occur due to sudden lane changes. Moreover DRL-based models were employed for real-time decision making of collision avoidance in (Cui et al. 2024; Wang et al. 2024; Wu et al. 2023). The system trained an agent to control brake inputs, make lane changes and adjust speed according to a given dynamic traffic scenario. DRL demonstrated superior performance compared to traditional rule-based systems, consistently outperforming and learning from new traffic scenarios. In studies (Khan and Ahmed 2023; Pang et al. 2024; Siems-Anderson 2024), the authors investigated the impact of adverse weather conditions (rain and fog) on traffic safety. They proposed an augmented version of the XGBoost model that includes weather data to determine the severity of the accidents with greater accuracy. It has been demonstrated that the inclusion of weather-related variables enhanced the accuracy of the prediction remarkably, providing a more consistent CAS in unfavorable weather conditions.

In studies (Naeem et al. 2025; Li et al. 2024, 2025), a hybrid model that combines recurrent neural networks (RNNs) and fully connected neural networks (FCNNs) was introduced to predict lane-changing behaviour. This system utilised past data and current sensor feedback to forecast the probability of a vehicle changing lanes, thereby allowing time to prevent a collision proactively. In (Tiwari et al. 2025; Na et al. 2024), the authors utilise FL to enhance the security of vehicle-to-vehicle communication, thereby preventing malicious attacks (such as false position) that could disrupt communication. Additionally, in (Arya et al. 2024; Civik and Yuzgec 2023), a cloud-based architecture was employed to identify faces and detect driver fatigue. The system utilised DL models, including VGG, ResNet and

CNN, to identify drowsiness in real-time. In studies (Çeven and Albayrak 2024; Vijayan et al. 2024), the authors introduced an ensemble ML framework comprising multiple layers designed to prevent road accidents. The architecture incorporated various ML models, specifically KNN, Naive Bayes and Decision Tree, to determine the occurrence of traffic accidents using diverse factors like traffic density and the nature of roads.

Several research studies have been conducted to explore the use of FL in replacing centralised learning schemes for collision avoidance in IoV systems, aiming to preserve privacy, reduce communication overhead and enhance model performance in real-time scenarios (He et al. 2024; Husnool et al. 2023; Thakur et al. 2025; Tabatabaie and He 2024). FedPPO (Federated Proximal Policy Optimisation) was employed to optimise reinforcement learning (RL) models for real-time decision-making in collision avoidance situations (Chen et al. 2023; Ghane et al. 2025; Gao et al. 2024; Yang et al. 2024). CNN-based drowsiness detection with FL-enhanced flexible adaptation of the system to changing driving conditions and ensured the privacy of the drivers (Zhang et al. 2022; Li, Zhang, et al. 2023). LSTM networks were applied to forecast vehicle paths, taking as input the time-series data, and FL methods were used to share updates among vehicles without information leakage about these sensitive data (Qiao et al. 2023; Min et al. 2024).

This research study highlights the potential of combining DL models with FL and digital twins to address the complex challenge of collision avoidance in IoV scenarios. However, significant gaps remain, especially in incorporating vehicular heterogeneity and human factors (e.g., driver fatigue, distractions) into such systems. The proposed study seeks to further investigate the conditions by integrating such factors into an FL-enabled CAS, with the use of a range of DL models, including CNN, LSTM, DRL and MLP, as well as GNN and digital twins to develop a more powerful, adaptive and privacy-preserving intelligent transportation system container. This study aims to enhance the safety and efficiency of IoV and mitigate the identified shortcomings of previous research studies by applying several FL approaches, including FedAvg, FedSGD, FedLU, FedGC and FedPPO.

3 | Proposed Methodology

In this section, the system architecture of our proposed FL-driven DL Models for Collision Avoidance in IoV, enhanced by digital twins integration, is discussed. This section presents methods for data collection, preprocessing, cleaning, feature selection and the merging of input parameters. It also outlines a proposed set of rules for data implementation and various algorithms. Our proposed work involves fixing the sensors on the vehicles to capture both our input parameters and the surroundings, as well as integrating digital twins to model the real-time behaviour of vehicles and the environment. Digital twins accurately reflect real-world conditions. We undergo a thorough calibration process. Each digital twin is tested and compared with actual data from vehicle sensors and telemetry. For example, we use driving data from various types of vehicles, like trucks, sedans and motorcycles, to simulate how these vehicles perform in different situations, such as varying

speeds, road types and weather conditions. In our approach, digital twins enhance Federated Learning by simulating real-time vehicle and environmental conditions. These virtual models help us better understand dynamic traffic scenarios and make more accurate predictions. Unlike traditional methods, digital twins allow for simulating complex conditions like traffic, weather and road surface. This integration enhances decision-making, ensuring safer and more adaptive collision avoidance in real-time. In our proposed intelligent system, the DL algorithms are adapted to determine whether an accident is going to happen or not according to the parameters gathered by the sensor and the digital twin models. The sensors will capture information concerning vehicle speed, the length of the vehicle, the weight of the vehicle, the distance between two vehicles, the humidity of the air, the road condition (i.e., wet, dry or snowy road), time and the condition of the driver calculated by the sensors based on the driver's exhalation in the air. These parameters, along with the real-time simulations from digital twins, are vital in forecasting accidents. In our proposed system, we select the dataset in real-time based on the proposed algorithms and digital twins data. We then train the model according to traffic rules, considering permissible vehicle behaviours, weather conditions and human representation. Our proposed architecture contains six steps as shown in Figure 2. These steps are data collection, data preprocessing, data sampling, the proposed algorithm, DL algorithms and model prediction.

During the implementation of a DL algorithm, we have explored various approaches to identify the most suitable algorithm in terms of accuracy for our proposed data-driven vehicular heterogeneity-based intelligent CAS in IoV. The characteristics of the problem determine the choice of DL algorithms in a research study, the extracted data characteristics, the computational resources and the performance evaluation implementation. Choices are made based on several factors, including the nature of learning (supervised), the structure of the dataset, interpretability needs, resistance to outliers, library availability and the results of trial-and-error testing and measurement. The operational workflow of the proposed FL-driven DL system of collision prediction in heterogeneous IoV environments integrated with digital twins is illustrated in Figure 2. This process begins with data collection and preprocessing, which involves cleaning and standardising the dataset. Subsequently, the data were sampled based on conditions such as vehicular heterogeneity, weather, HB and real-time simulations provided by digital twins.

Several DL models, including CNN, LSTM, MLP, DRL and GNN, are chosen and coupled with more customised FL approaches, including FedAvg, FedSGD, FedLU, FedPPO and FedGC. The models allow distributed learning among vehicles, without sharing raw data and minimising communication overhead. The examination of the prediction results is based on measuring performance using precision, recall, F1-score and support to identify the most efficient model for a real-time and privacy-preserving collision avoidance situation. Figure 3 demonstrates the architecture of a system based on FL in an IoV environment, including data from digital twins. The Global Server receives model updates from various FL Clients that are linked to Roadside Units (RSUs). The

RSUs store and process data locally with their own AI model, which is constantly updated by the interaction of vehicles and the surrounding environment. These updates are sent to the local models, enhancing the FL AI Model as they are aggregated, while maintaining privacy and reducing communication overhead. Figure 3 illustrates the diversity of transport (car, tractor, bus, truck and bike) in conversation with their RSUs, showcasing the dynamic performance of an intelligent transportation system.

3.1 | Dataset Collection

Data collection is the initial stage of our proposed system architecture. The choice of suitable datasets or sensors used in the vehicular collision system is crucial, as it directly impacts the system's overall design and, consequently, its outcome. The architecture of our system is quite flexible, as it supports a large number of different datasets that can be applied in the sphere of vehicular collisions. In our case, we rely on publicly available real-world solutions to collision prediction. The dataset (Onesimu et al. 2021) provides data of different parameters of vehicular heterogeneity, such as the influence of weather conditions and the conditions of the road surface on driving processes. The dataset comprises 311,908 observations, representing a range of driving conditions in real-world scenarios and is heterogeneous in terms of vehicle categories and road conditions. Moreover, Dataset (Tao et al. 2024; Li et al. 2022) also concerns HBF, including driver fatigue, distractions and other factors related to behaviour that may change the decisions when driving. There are more than 100,000 values in Dataset (Tao et al. 2024) and about 3000 values in Dataset (Li et al. 2022). To evaluate our models properly, we split the data into three parts: 70% for training the models, 15% for validating and fine-tuning the models and 15% for testing the model's performance. This way, we make sure the model is trained on one set of data, tuned on another, and finally tested on data it hasn't seen before, ensuring a fair evaluation and avoiding any data leakage. These datasets will be fundamental in modelling human behavioural components into the prediction model of collision since HB in most cases contributes immensely to accidents. The composition of these datasets enables the system to simulate a wide range of real-world situations, thereby enhancing the precision and trustworthiness of our FL-based collision avoidance models. Data has been processed and preprocessed to ensure its suitability for DL applications, real-time predictions and adaptive CASs that incorporate environmental dynamics and behavioural factors.

3.2 | Data Preprocessing

Preprocessing of data is critical because it helps to clean the data, determine and correct errors, including noise, missing data, and outliers, and normalise the data. Since the amount of outliers and noise in the data significantly affects the accuracy of the DL model, the initial step is to eliminate these values from the dataset. In Dataset (Onesimu et al. 2021), the missing values have been substituted by an average of the parameters they represented after computing each of the values. There is a significant amount of noisy data, along with a substantial

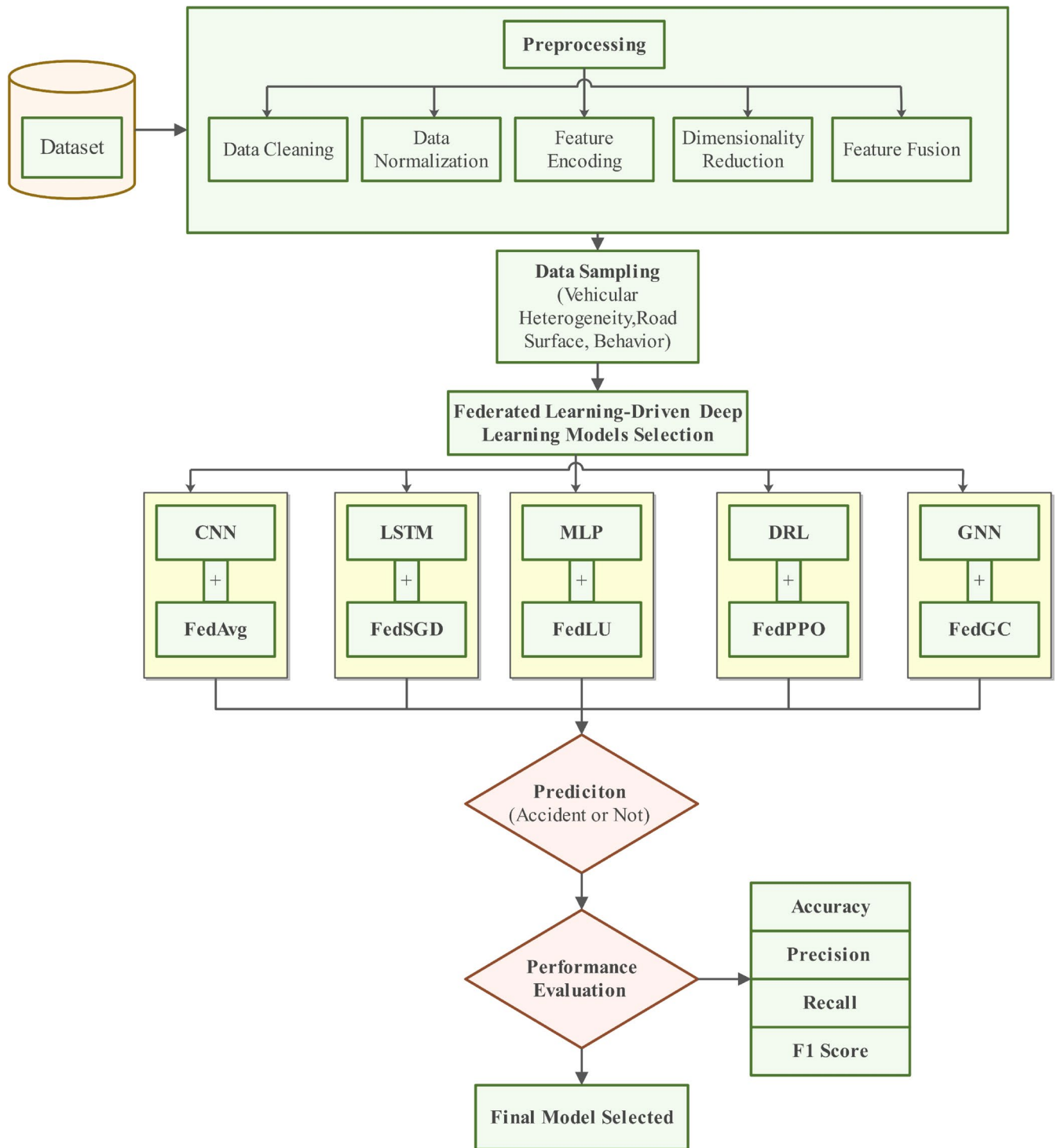


FIGURE 2 | Flowchart of proposed study.

number of missing values, which were addressed to achieve a reasonable level of accuracy. We use DL-based normalisation, such as Batch Normalisation, to process the data and achieve a more consistent distribution on the scale. Batch normalisation helps scale the input functions before they enter the neural network, setting the mean value to zero and the variance to one for each layer, which in turn improves training and convergence. Moreover, Dataset (Tao et al. 2024) and Dataset (Li et al. 2022) are processed to produce the input values of interest that will be utilised to create the final dataset to be applied

to train the model. The parametric data in these datasets are reprocessed to achieve uniformity and eliminate any variance that can hinder the learning process. Another essential process of enhancing the DL model's successful level during training and actual application is Data Normalisation (DN). We consider the DN techniques, such as Layer Normalisation, to normalise the data. The method scales the features in the dataset appropriately, thereby enhancing the model's performance by ensuring that input features contribute equally to the training process.

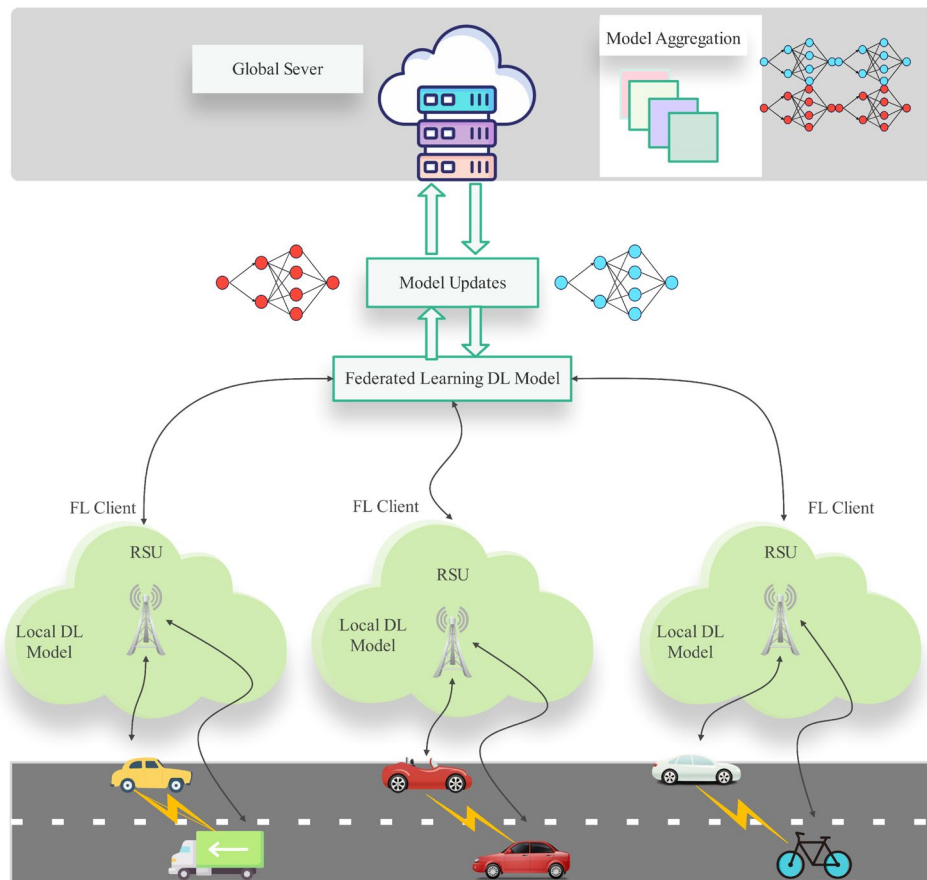


FIGURE 3 | Federated learning architecture.

3.3 | Data Sampling

The process of data sampling is a critical component that ensures that the dataset is categorised into various groups aimed at improving the speed and success of the corresponding DL algorithms. In our proposed research, we categorised the Dataset (Onesimu et al. 2021) into 12 various categories of Road Surface Status (RSS) and heterogeneity of vehicles. Such a classification facilitates the simplification of the training process, as well as the adequacy and organisation of the data fed into the DL models. When the values in the dataset are highly correlated with each other, the prediction system becomes more accurate. When a strong correlation does not exist between the values in a dataset, additional preprocessing techniques are applied to enhance the quality and suitability of the dataset for training. The DL-based correlation technique is used to build a model for correlation analysis, which helps refine the dataset and enhance the prediction rates of the proposed CAS. This is improved by using the best sampled and correlated data in model training, resulting in a higher level of accuracy in predicting and avoiding collisions.

3.4 | Aggregate Strategies in Federated Learning

The aggregation step in FL combines locally trained models from different clients (e.g., vehicles or Roadside Units) to form a better global model. To keep everything in sync, each vehicle first trains its model locally for a set number of epochs before

sending its updates to the central server. Synchronisation happens at regular intervals, typically every $\langle \text{time period} \rangle$ (e.g., every 30s), to update the global model. If there are fewer vehicles on the road or if the network connection is unstable, synchronisation may be delayed to prevent overloading the system and to improve communication efficiency. Various aggregation strategies have been proposed to accomplish this, all achieving different advantages and trade-offs in terms of accuracy, efficiency and preserving privacy. The major aggregation strategies of FL are presented below, with their mathematical definitions and descriptions.

3.4.1 | Federated Averaging

The FedAvg algorithm updates the global model by averaging the model updates (parameters or weights) obtained from all local models. We are dealing with sensitive user data; we've built in strong privacy protections to keep personal information safe. To start, we use secure aggregation, which ensures that only encrypted updates are sent to the central server. This way, no individual vehicle's data is ever exposed. Additionally, we apply differential privacy during model training, introducing a level of 'noise' to the updates, thereby preventing anyone from tracing any specific vehicle's data back to the model. By combining these two methods secure aggregation and differential privacy we can safeguard privacy while still allowing our system to make accurate predictions, all while keeping the data decentralised and secure. In the Equation (1), w_{t+1} represents the updated global model after the

$(t + 1)$ th iteration and $w_i^{(t)}$ represents the model parameters from the i -th client at the t th iteration. In this case, N is the number of clients (e.g., vehicles or RSUs) (Rajagopal et al. 2023).

$$w_{t+1} = \frac{1}{N} \sum_{i=1}^N w_i^{(t)} \quad (1)$$

Every client calculates their model and shares the new weights with a central server. The server averages these updates to create a more accurate global model, and the process begins again in the next round. According to this approach, all clients contribute equally to the global model; thus, the aggregation is simplified, and privacy is ensured since no raw information is sent.

3.4.2 | Federated Stochastic Gradient Descent

In FedSGD, the aggregation is done by averaging the gradient updates rather than directly averaging the model parameters. Here in Equation (2) g_{t+1} represents the aggregated gradient at the $(t + 1)$ th round, and $g_i^{(t)}$ is the gradient computed by the i -th client at the t th round (Xu et al. 2023).

$$g_{t+1} = \frac{1}{N} \sum_{i=1}^N g_i^{(t)} \quad (2)$$

This method addresses the issue of averaging model parameters by focusing on averaging the gradient information (i.e., the direction of change), which can be more effective with large datasets. Once aggregated, the weights in the global model are updated with the average of the gradients of all clients, reducing the communication overhead and total workload on the server without revealing any information.

3.4.3 | Federated Proximal Policy Optimisation

FedPPO is an aggregation strategy using reinforcement learning. It stabilises the training process by avoiding too large updates. In the Equation (3) θ_{t+1} represents the model parameters at the $(t + 1)$ th iteration, and θ_t represents the model parameters at the t th iteration. The term α is the learning rate, and $\hat{A}_t$ is the advantage function, which helps the agent (vehicle or RSU) determine whether an action is better than the baseline (Khan et al. 2024).

$$\theta_{t+1} = \theta_t + \alpha \cdot \text{clip}(\hat{A}_t, -\epsilon, \epsilon) \cdot \nabla_{\theta} J(\theta) \quad (3)$$

The clipping term ϵ ensures that the updates are not too drastic, contributing to stability. Finally, $\nabla_{\theta} J(\theta)$ is the gradient of the objective function, directing the update. This strategy enables informed decision-making in dynamic environments like IoV by maximising cumulative rewards while minimising abrupt model changes.

3.4.4 | Federated Graph Convolution

In FedGC, the global model is updated by aggregating information from neighbouring nodes (vehicles or RSUs) in a

graph-based structure. Here in Equation (4) $h_v^{(t+1)}$ represents the updated feature vector for node v (vehicle or RSU) at the $(t + 1)$ th iteration, and $N(v)$ represents the set of neighbouring nodes (Adnan, Syed, et al. 2025).

$$h_v^{(t+1)} = \sum_{u \in N(v)} W \cdot h_u^{(t)} + b \quad (4)$$

W denotes the weight matrix of the convolutional layer and b is the bias term. Each node in FedGC can modify its representation using data collected by neighbours to describe the interactions between two or more vehicles or parts of the infrastructure within an IoV system. This assists in creating a more cooperative model by incorporating the communication of vehicles in the network to enhance forecasts of vehicle actions and collision avoidance.

3.4.5 | FL With Local Update

In FedLU, each client performs several local updates before sending its model parameters to the central server. Here in Equation (5) $\theta_i^{(t+k)}$ represents the parameters of the model for the i -th client after k local updates, and $\theta_i^{(t)}$ represents the model parameters at the t th iteration (Wu et al. 2022).

$$\theta_i^{(t+k)} = \theta_i^{(t)} - \eta \cdot \nabla L_i(\theta_i^{(t)}) \quad (5)$$

The learning rate η controls the step size of the updates, and $L_i(\theta_i^{(t)})$ is the local loss function for the i -th client, measuring how well the local model performs on its data. This approach has the advantage of allowing each client to apply local updates before synchronising, so that the locally trained models are refined to the dataset specifics of that client, which improves the performance of the global model once aggregated. This is particularly useful when data is skewed across clients.

3.5 | Deep Learning Models

Various DL models are utilised and developed within the framework of this study to ensure collision avoidance in the IoV. DL is very successful in describing and learning complex patterns and representations, especially when using multi-layered artificial neural networks with large-scale data. Such models are carefully selected to handle all types of data obtained from the vehicle, sensors in the environment and HB. The models used in this study are as follows:

3.5.1 | Convolutional Neural Networks

In this study, CNNs are utilised to process data collected by onboard cameras that detect possible obstacles, vehicles on the road or pedestrians. The CNN consists of several layers: convolutional layers filter the image to extract useful features, pooling layers downsample the image, and fully connected layers provide an output signal indicating the decision to avoid collision (Muzahid et al. 2022). Such a model enables the system to identify and sort objects in real-time, allowing it to forecast possible collisions in advance using visual input.

3.5.2 | Long Short-Term Memory

LSTM networks are a long short-term memory neural network that works with sequential-related data, as it is common in time-series data, e.g., vehicle trajectories. LSTMs can store valuable information over time through the memory cells within each LSTM, enabling long-term connections to form within the data. The model utilises LSTM layers to gain insight into future vehicle movements by analysing past movement data. A dense layer with only one LSTM layer in our system gives a probability of collision, considering the predicted future position of vehicles (Khan and Ahmed 2023). The LSTM layer activation function is Rectified Linear Unit (ReLU) and the dense layer activation function is sigmoid, which returns a binary classification output related to collision prediction.

3.5.3 | Deep Reinforcement Learning

DRL enables vehicles to make real-time decisions on how to avoid collisions, such as braking, accelerating or steering. Acting in a dynamic environment, the agent learns optimum actions using trial and error, as rewards or penalties are given to the actor. The policy of the agent is typically optimised through Deep Proximal Policy Optimisation (PPO), which facilitates stable learning. The DRL agent receives the environment state (car position, velocity and traffic) as input and returns some actions (Ullah et al. 2023). In this way, the model can continuously respond to changing traffic conditions. The agent becomes more proficient at collision avoidance over time. The reinforcement learning model ensures a trade-off between exploration and exploitation to make the most efficient decisions. It streamlines its policy through continuous feedback, ensuring accuracy. The ability of DRL to make real-time dynamic predictions and prevent collisions is essential to IoV systems.

3.5.4 | Graph Neural Networks

In GNNs, the neural network uses the graph structure to model the relations and interactions between multiple vehicles or objects within the IoV system. The nodes describe each vehicle, whereas the edges relate the nodes and can describe either an interaction or a dependency, such as proximity or the movements of a vehicle. GNNs combine data of near nodes, which lets the model learn complicated relations in vehicle behaviours. Since the model can capture both spatial and temporal patterns using graph convolutions, it is capable of foreseeing how the vehicles will interact with each other in the future and whether any collisions will occur (Pang et al. 2024). The graph convolutional layer aggregates the information of adjacent nodes, whereas the dense layer, with a sigmoid activation function, returns the predicted collision. This method provides further collision avoidance through modelling to support the interaction of vehicles within a network and aid in a dynamic environment.

3.5.5 | Multilayer Perceptron

MLP is a FCNN that is used in our model to process structured data such as vehicular features (e.g., speed, weight, size) and

environmental factors (e.g., road conditions, weather). MLPs consist of an input layer, one or more hidden layers and an output layer. The nonlinear activation functions of the hidden layers are the ReLU, which represents complex relationships among the input features and the output layer has a sigmoid activation function that predicts the probability of a collision (He et al. 2024). This model provides an excellent approximation of the collision risk, which is dependent on various variables inherent to the vehicles and environmental conditions.

To overcome the challenges of preserving privacy, scalability and real-time decision making, DL models are combined with FL and digital twins. FL allows the training of models to be decentralised whereby each vehicle trains its local model on its data and transfers the model updates to a central server. This method minimises the need to share raw data, thereby preserving the confidentiality of sensitive information about the vehicle's location, speed and preventive practices. The server then combines the model updates received by all vehicles and digital twins and the global model is further optimised and sent back to all participating vehicles. This is an iterative process where the models become more accurate each time, ensuring they perform to the best of their abilities and adapt to continuously changing road conditions. Our system successfully predicts potential collisions in real time, leveraging the combination of DL models, FL and digital twins to enhance privacy, scalability and decision-making capabilities across a vast array of vehicles in the IoV network.

4 | Experimental Setup

In this section, the basic elements and necessities of the proposed FL-based CAS in the IoV with digital twins integration are described. It includes the values of the datasets involved, the sensors applicable for data capture, and the system requirements that are to be implemented and tested to facilitate the success of the proposed models in application and assessment. In our research, we examine the implementation and evaluation of DL models based on FL, which we carry out through the optimisation of the Flower framework on Google Colab. Flower is oriented on big-scale FL deployment and may support machine and DL frameworks such as Keras and PyTorch, preparing it to suit the requirements of real-world IoV sets characterised by a sizable number of clients (e.g., vehicles and Roadside Units, RSUs). In FL, clients execute local computations with actual data on their nodes. In contrast, the global computation is conducted at a centralised server that aggregates the model updates from the clients. Such decentralisation does not reveal raw data but rather shares updates on a model. The architecture of Flower enables interaction between local and global computations, enhancing the model's accuracy after multiple training iterations with digital twins and further optimising the system's real-time adaptability.

4.1 | Dataset

Our experiments are based on a publicly available dataset, on which we train our CAS, driven by FL within the IoV with digital twin integration. Our dataset is the one presented

TABLE 2 | Human behaviour attributes for driver monitoring in collision avoidance systems.

Attribute	Range	Data type	Description
Alcoholic	Alcoholic, Not_Alcoholic	Floating point	Indicates whether the driver is alcoholic or not.
Drowsiness	Eyes Open, Eyes Closed, Yawn, No Yawn	Floating point	Indicate the driver's state of alertness (eyes open/closed, yawning).

by Onesimu et al. (2021), which provides vehicle and RSS (Norway) data collected over two years. Furthermore, we also use HB datasets, such as the alcohol influence data set (Tao et al. 2024), which follows the driving behaviour under alcohol and the drowsiness detection data set (Li et al. 2022), which examines the drowsy character of the driver in terms of eye as well as yawn detection. A dataset selection algorithm is applied to these datasets to extract the best from both long and short vehicle data, depending on road conditions and vehicle type. This enables the system to learn from a wide variety of information sources, thereby enhancing the robustness and accuracy of collision avoidance predictions in dynamic IoV settings. Table 2 lists the attributes to be employed in modelling CASs and observing driver behaviour. Table 2, in turn, lists the characteristics of HB, that is, the status of alcohol consumption and the degree of alertness of the driver himself, which is identified based on some manifestations of drowsiness, such as eye movements or yawning. These features are essential for determining the chances of the driver responding and ensuring the proper functioning of a collision-avoiding system. Collectively, these Tables provide an all-inclusive method for safe driving, as they incorporate vehicle environmental information and HB cues.

4.2 | Sensors for the Federated Learning-Driven CAS in IoV

In our proposed FL-based CAS for the IoV with digital twins integration multiple sensors will be required to provide real-time information necessary for making accurate predictions about potential collisions. These are speed sensors that measure the vehicle's speed. This camera sensor detects road conditions, including harmful HB such as driver drowsiness or distraction. To consider environmental conditions, we suggest installing humidity and temperature sensors to monitor road weather conditions, a rain sensor to measure precipitation levels, and an ultrasonic sensor to detect the presence of objects around the vehicles. There will also be an alcohol sensor (MQ-3) employed to determine the concentration of alcohol in the driver's breath, as well as visibility sensors that will help assess road visibility. Whereas we train the system in our study using published datasets, these sensors will be applied to the systems in future real-life situations to gather dynamic, real-time information.

5 | Results and Discussion

The results of collision prevention using several DL models combined with FL in the IoV with digital twins integration are given in this section and compared to traditional approaches. We utilise various models, including CNN, LSTM, DRL, MLP

and GNN, on real-world IoV datasets. The metrics applied to assess the models are accuracy, precision, recall, F1-score and confusion matrix. The comparison of the efficiency of each model in predicting and preventing crashes is conducted, with particular emphasis on the role of vehicle heterogeneity, environmental conditions and other factors related to HB. Our proposed system works by using real-time data from V2V and V2I communication. It takes into account the differences between vehicles, including their size, speed and the types of sensors they utilise. For instance, larger vehicles like trucks typically take longer to respond compared to smaller vehicles like sedans, which can increase the risk of collisions. To address this, we utilise GNNs, which excel at understanding how vehicles interact with one another on the road and can adapt to the unique behaviour of each vehicle type. This helps improve collision avoidance, even when there's a wide variety of vehicles on the road. Additionally, we utilise Digital Twins, virtual models of real vehicles and road conditions, to enhance the accuracy of our predictions. These twins simulate how different vehicles perform under various conditions, particularly in scenarios where multiple types of vehicles interact. This enables our system to make decisions in real-time, helping to ensure safer interactions between all kinds of vehicles. Moreover, the findings are also contrasted with various FL algorithms, namely FedAvg, FedSGD, FedPPO, FedLU and FedGC, to get a clear picture of how the proposed approach gains advantages over the others in terms of scalability, privacy, and absence of waiting time to make real-time decisions. Table 3 and Table 4 contain information about vehicle features, environmental conditions, and accidents and Table 4 also includes supplementary features of HB. Table 3 contains vehicle characteristics, such as length, speed and weight, as well as lead vehicle characteristics and environmental factors, including relative humidity, precipitation, road surface conditions and timing. It also indicates whether an accident occurred or not. This is further developed in Table 4, which considers more specific human factors, such as the driver's drowsiness (as indicated by open/closed eyes and yawning) and the level of alcohol, as well as the time lag between vehicles and the length of the lead vehicle. The purpose of both tables is to model collision risks and accidents. Still, Table 4 has a more extensive dataset, as it incorporates most critical data driver-related, such as the type of driver, number of accidents and the police involved in the collision, and so forth, which makes it more adequate for optimising collision avoidance and driver monitoring systems.

5.1 | CNN+ FedAvg Model Result

The CNN + FedAvg model trains on a real-world dataset that combines variables such as vehicle speed, RSS, HB, driver fatigue and distractions. Table 5 shows the classification report of

TABLE 3 | Values from dataset (Onesimu et al. 2021).

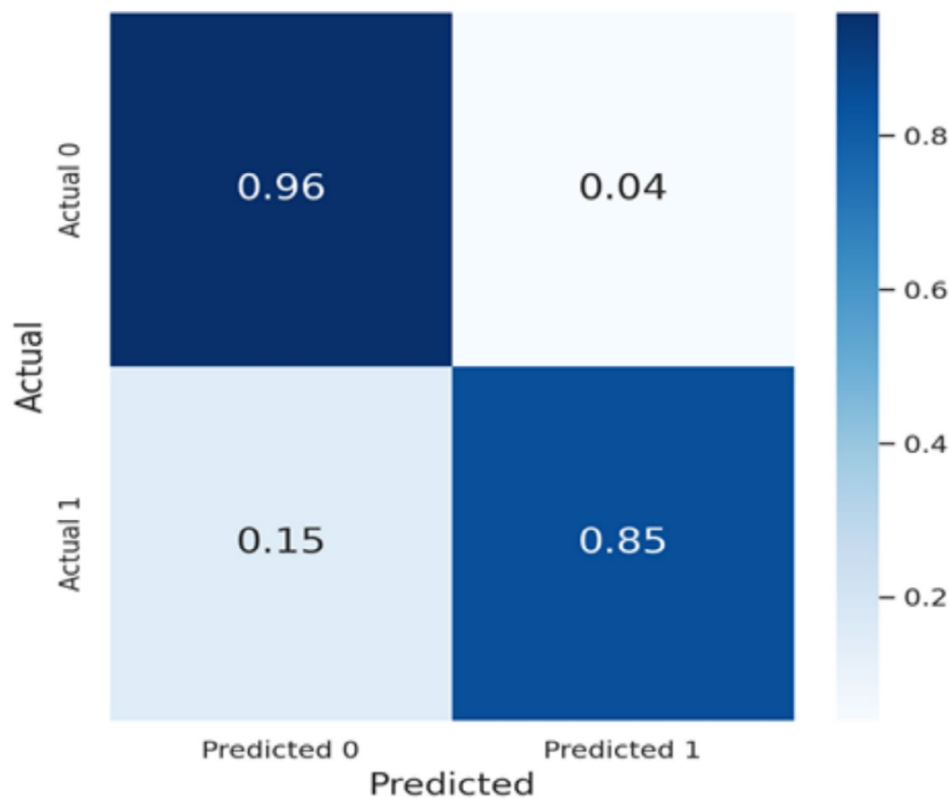
ID	V-length	Lane	V-speed	V-weight	L-V-speed	L-V-weight	Time gap	Relative humidity	Precipitation type	Road surface status	Time of day	Accident
1	1708	2	82	24,695	84	1717	449	87	Snow	Dry	Daylight	1
2	1551	3	83	2205	82	24,695	513	87	Rain	Dry	Twilight	0
3	1428	1	76	13,725	78	1437	527	87	Rain	Dry	Daylight	0
4	1591	3	83	2205	85	24,695	518	87	Rain	Dry	Daylight	1
5	524	2	87	1797	95	2075	513	87	Snow	Snow covered	Twilight	0

TABLE 4 | Values from generated dataset.

ID	V-length	Lane	V-speed	V-weight	L-V-speed	L-V-weight	L-V-length	Time gap	RH	Precip. type	Road surface	Time of day	Drowsy	Alcoholic	Accident
1	1708	2	82	24,695	84	1717	449	2,357,571	87	Snow	Dry	Daylight	Eyes close	Alcoholic	1
2	1551	3	83	2205	82	24,695	513	5,520,146	87	Rain	Dry	Twilight	Eyes open	Not Alcoholic	0
3	1428	1	76	13,725	78	1437	527	174,896,739	87	Rain	Dry	Daylight	Eyes open	Not Alcoholic	0
4	549	3	83	17,365	84	1563	527	3,465,379	87	Rain	Dry	Daylight	Yawn	Not Alcoholic	0

TABLE 5 | Comparison of classification reports for CNN + FedAvg on two datasets.

Class/metric	Dataset (Onesimu et al. 2021)			Generated dataset		
	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.93	1.00	0.96	0.96	1.00	0.98
1	0.98	0.84	0.90	1.00	0.85	0.92
Accuracy		96.5%			97%	
Macro avg	0.96	0.92	0.93	0.98	0.92	0.95
Weighted avg	0.94	0.93	0.94	0.97	0.96	0.97

**FIGURE 4** | Confusion matrix for CNN + FedAvg on generated dataset.

the CNN + FedAvg model on the dataset (Onesimu et al. 2021) and the generated dataset.

The CNN + FedAvg model used in this study achieves an accuracy of 97% in predicting collision risks using the generated dataset, reflecting its high predictive performance. The confusion matrix of the CNN + FedAvg model on the produced dataset shows that 85% of actual positive cases were correctly detected as positive (true positive rate) and 96% of actual negative cases were correctly detected as negative (true negative rate). Furthermore, 15% of the real positive candidates were calculated incorrectly as low (false negative rate) and only 4% of the real negative candidates were selected as high (false positive rate). These values indicate that the CNN + FedAvg model has excellent accuracy in recognising both the positive and negative instances, as shown in Figure 4. Notably, the integration of digital twins in this model enabled a more accurate simulation of real-time vehicle and environmental conditions, contributing to the model's enhanced ability to predict collision risks.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (6)$$

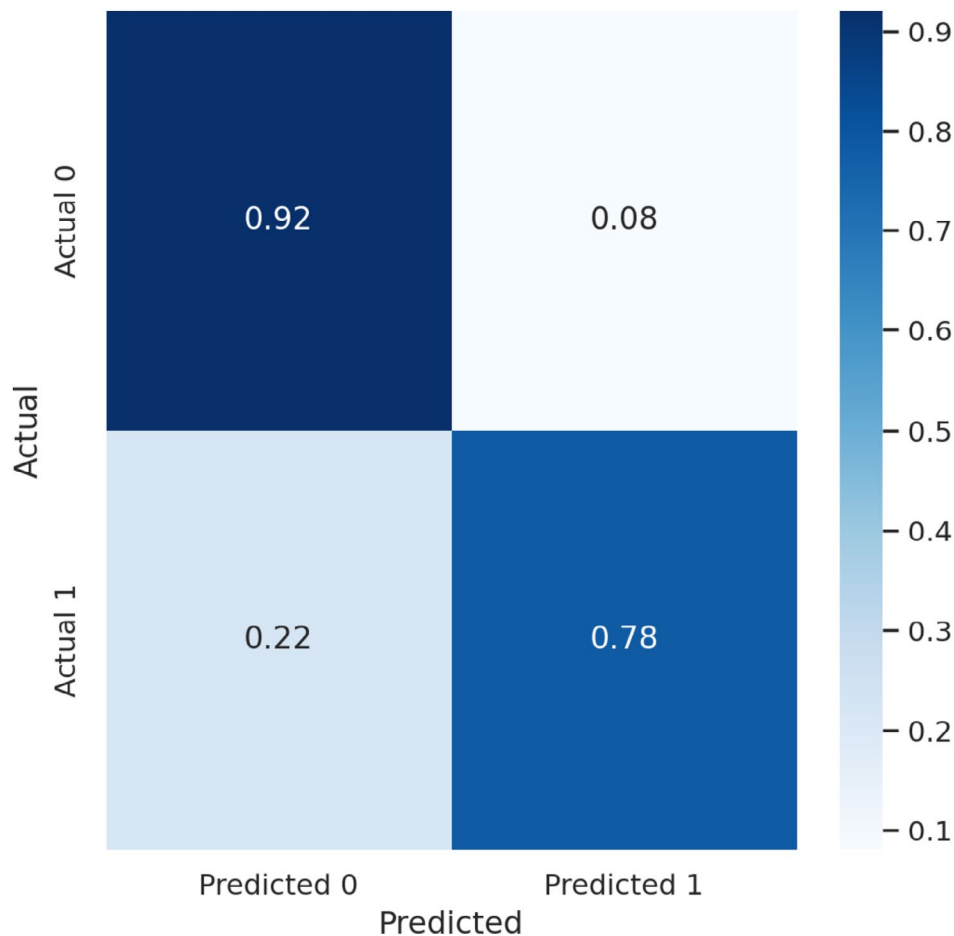
Table 5 presents the precise performance metrics of the CNN + FedAvg model, including precision, recall and F1-score measures, which demonstrate the model in real-time collision prediction. The findings demonstrate the power of the CNN + FedAvg combination to provide high accuracy, privacy-friendly and real-time collision avoidance forecasting, enabling a safer road in dynamic and decentralised vehicular networks.

5.2 | LSTM + FedSGD Model Result

We have applied LSTM, a kind of RNN, combined with the FedSGD FL method to solve the collision avoidance issue in the IoV. The integration enables decentralised training, resulting in enhanced data privacy and reduced communication overhead between participating vehicles.

TABLE 6 | Comparison of classification reports for LSTM + FedSGD on two datasets.

Class/metric	Dataset (Onesimu et al. 2021)			Generated dataset		
	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.92	1.00	0.96	0.94	1.00	0.97
1	1.00	0.78	0.87	1.00	0.83	0.91
Accuracy		97.2%			97.65%	
Macro Avg	0.96	0.89	0.91	0.97	0.92	0.94
Weighted Avg	0.94	0.93	0.93	0.95	0.95	0.95

**FIGURE 5** | Confusion matrix for LSTM + FedSGD on the generated dataset.

The LSTM + FedSGD model was also tested on the generated dataset in our study and achieved a remarkable accuracy of 97.65%, as presented in Table 6. Figure 5 depicts the confusion matrix of LSTM + FedSGD on the synthetic data, with a true positive rate (TPR) of 78%, a true negative rate (TNR) of 92%, a false negative rate (FNR) of 22% and a false positive rate (FPR) of 8%. Table 6 displays the performance measures of the LSTM model combined with FedSGD, which were obtained by running the model on the generated dataset, demonstrating the model's proficiency in collision prediction. The results demonstrate the effectiveness of the LSTM + FedSGD model, which is highly capable of learning long-range temporal dependencies and making accurate real-time decisions in the IoV environment, considering data privacy provided by distributed sources. Furthermore,

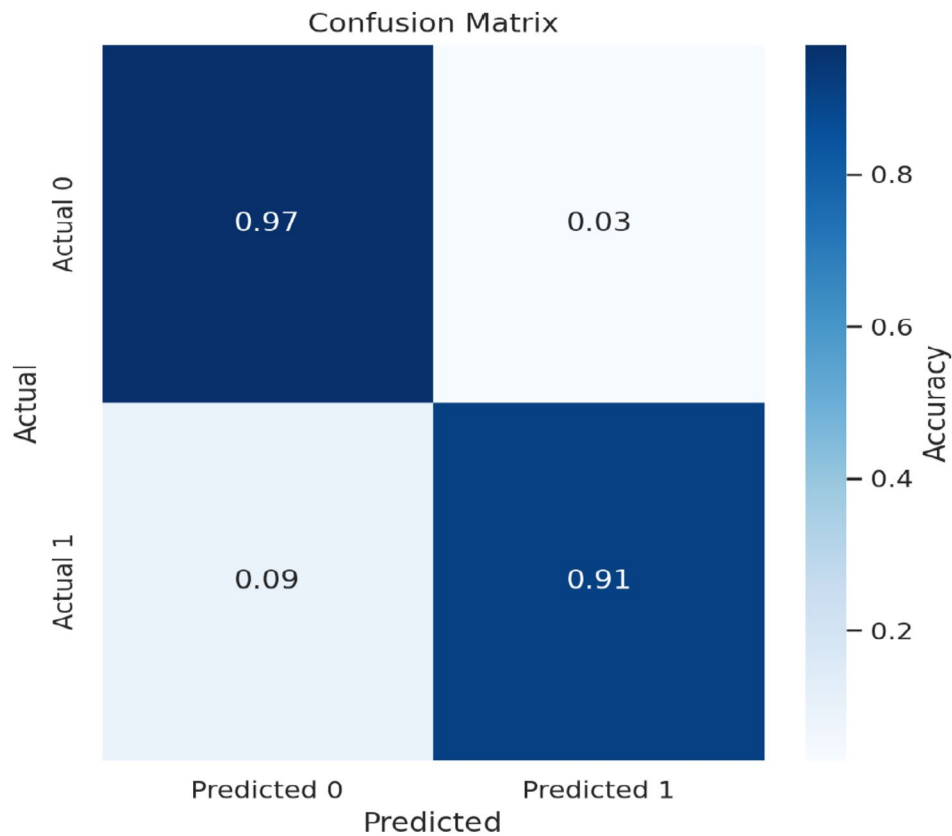
the integration of digital twins further enhanced the model's performance by simulating dynamic traffic conditions and vehicle behaviour in real-time.

5.3 | Deep Reinforcement Learning (DRL + FedPPO) Model

We used DRL with FedPPO to handle the collision avoidance issue in the IoV in the proposed study. This composition enables agents (vehicles) to learn the optimal policies in a decentralised way and keep the privacy of data and minimise communication overhead. DRL + FedPPO is well suited to dynamic settings where real-time decision making and adaptation are essential.

TABLE 7 | Comparison of classification reports for DRL + FedPPO on two datasets.

Class/metric	Dataset (Onesimu et al. 2021)			Generated dataset		
	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.97	1.00	0.98	0.97	1.00	0.98
1	1.00	0.90	0.95	1.00	0.91	0.95
Accuracy		98.0%			98.3%	
Macro Avg	0.98	0.95	0.97	0.98	0.95	0.97
Weighted Avg	0.98	0.98	0.98	0.98	0.98	0.98

**FIGURE 6** | Confusion matrix for DRL + FedPPO on the generated dataset.

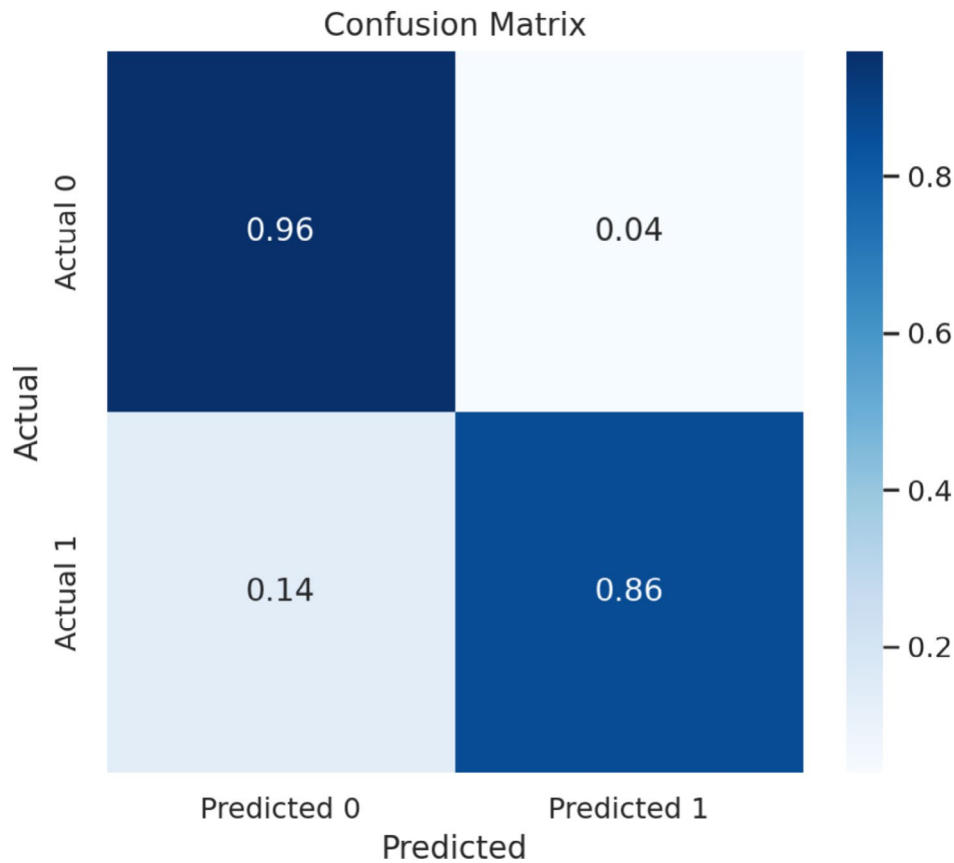
The generated dataset was tested with the DRL + FedPPO model, achieving an incredible accuracy of 98.3%, as depicted in Table 7. In Figure 6, the confusion matrix of DRL on the generated dataset indicates that 91% of actual positive cases were correctly classified as positive (true positive rate), whereas 97% of actual negative cases were correctly labelled as negative (True Negative Rate). In the meantime, the FNR was 9% and the FPR was 3%. The results, combined with the performance measures described in Table 7, emphasise the effectiveness of DRL, combined with FedPPO, in real-time, decentralised collision prediction and collision avoidance. The integration of digital twins played a key role in improving the model's ability to adapt to dynamic traffic scenarios. By providing real-time simulations of vehicle and environmental conditions, digital twins enhanced the model's accuracy in predicting potential collisions. This also positively affects model accuracy and enables privacy-aware, adaptable decision-making in various traffic situations within the IoV ecosystem.

5.4 | MLP + FedLU Model

The multilayer perceptron (MLP) combined with FedLU, a supervised DL algorithm, was employed in the proposed study to solve the collision avoidance problem in the IoV. An accuracy of 98.5% was obtained when the MLP + FedLU model was applied to the generated dataset, as illustrated in Table 8. The corresponding TPR is 86%, true negative rate (TNR) is 96%, FNR is 14% and FPR is 4% as in the confusion matrix in Figure 7. The performance measurement of the implementation is explained in Table 8, which validates the model to predict the collisions in vehicles. The findings indicate that MLP + FedLU, which possesses the capability to structure the relations between data points in a nonlinear form, provides a reliable, privacy-preserving and scalable decision-making methodology applicable to the real-time operations of IoV-based CASS.

TABLE 8 | Comparison of Classification Reports for MLP + FedLU on two datasets.

Class/metric	Dataset (Onesimu et al. 2021)			Generated dataset		
	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.95	0.99	0.97	0.96	1.00	0.98
1	0.97	0.83	0.89	0.98	0.86	0.91
Accuracy		97.8%			98.5%	
Macro avg	0.96	0.91	0.93	0.97	0.93	0.94
Weighted avg	0.96	0.94	0.95	0.97	0.95	0.96

**FIGURE 7** | Confusion matrix for MLP+ FedAvg on generated dataset.

5.5 | GNN + FedGC Model

In this proposed study, GNNs, integrated with FedGC, are utilised to address the collision avoidance problem in the IoV. The produced dataset was used in the GNN + FedGC model to achieve an impressive accuracy of 98.8%, as presented in Table 9. Figure 8 shows the confusion matrix of GNN + FedGC on the created dataset. Here, the TPR of this matrix is 92%, the TNR is 99%, the FNR is 8% and the FPR is 1%. As shown in Table 9, the model's performance measures are excellent in capturing complex inter-vehicle relationships and forecasting potential collisions with a high degree of accuracy. The integration of digital twins further enhances the GNN + FedGC model by providing detailed, real-time simulations of vehicle and environmental conditions. These virtual models enabled the system to simulate various traffic scenarios, including complex interactions

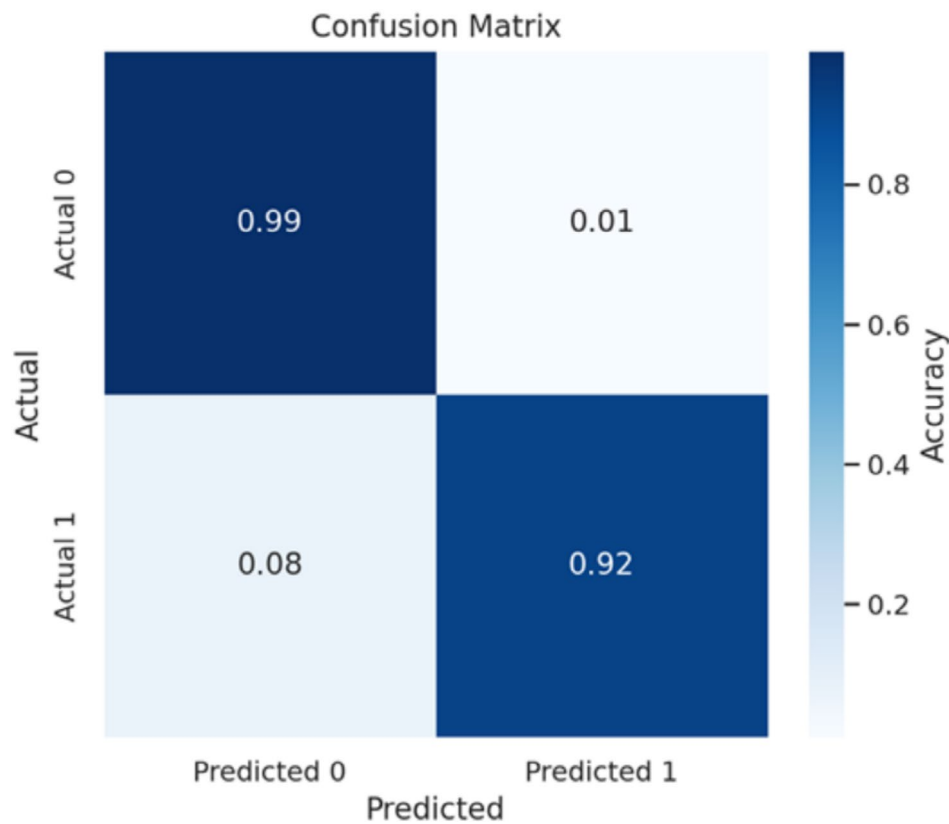
between multiple vehicles and predict collisions with greater accuracy. The GNN + FedGC model is adaptive and learns across decentralised nodes in IoV, offering strong privacy, which makes it an essential system component for real-time decision-making in smart cities, where high accuracy and responsiveness of model predictions are imperative.

5.6 | Performance Evaluation

In this section, we give a performance comparison of our proposed FL-based DL models with state-of-the-art in the area of collision avoidance in the IoV. We compare our method with the data-driven approach of Adnan, Umer, et al. (2025), which utilises a vehicular heterogeneity-based intelligent CAS in IoV. Our method is compared to the Traffic Accident Prediction System by Adnan,

TABLE 9 | Comparison of classification reports for GNN + FedGC on two datasets.

Class/metric	Dataset (Onesimu et al. 2021)			Generated dataset		
	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.98	1.00	0.99	0.99	1.00	0.99
1	1.00	0.90	0.95	1.00	0.92	0.96
Accuracy		98.5%			98.8%	
Macro avg	0.99	0.95	0.97	0.99	0.96	0.98
Weighted avg	0.99	0.98	0.99	0.99	0.98	0.99

**FIGURE 8** | Confusion matrix for GNN+ FedGC on generated dataset.

Umer, et al. (2025), which focuses on predicting accidents based on similar parameters, such as vehicle heterogeneity, weather and road conditions. In their study, Adnan et al. employed five ML algorithms: Support Vector Machine (SVM), Random Forest (RF), Naive Bayes, Logistic Regression and Decision Tree (DT) to quantify the performance of these models. We used four primary metrics: F1-score, Precision, Recall and Accuracy. Table 10 presents the performance measures of the existing systems, as reported by Adnan, Umer, et al. (2025), which were applied to the data (Onesimu et al. 2021). Comparatively, our proposed DL frameworks, including those based on CNN and FedAvg, LSTM and FedSGD, DRL, FedPPO, MLP, GNN and FedGC, outperform. Furthermore, the incorporation of digital twins in our models enhanced the simulation and prediction of real-time traffic conditions, vehicle interactions, and environmental factors, further improving the accuracy and adaptability of the system. These DL model results with plain dataset (Onesimu et al. 2021) and on a

TABLE 10 | Performance of the existing system.

Algorithm	Precision	Recall	F1-score	Accuracy
SVM	0.94	0.94	0.94	94%
Naïve Bayes (NB)	0.93	0.93	0.93	93%
Random Forest (RF)	0.93	0.93	0.93	93%
Decision Tree (DT)	0.90	0.92	0.91	90%
Logistic Regression (LR)	0.92	0.93	0.92	92.5%

generated dataset with HB as a factor are reported in Tables 11 and 12, respectively. The inclusion of digital twins significantly enhanced the model's ability to simulate complex real-world scenarios, resulting in more accurate and reliable collision predictions, particularly in dynamic environments.

Figure 9 shows a comparison of the accuracy of various ML models on the dataset (Onesimu et al. 2021). Based on the Figure 9, it is clear that the model with the SVM provided the best accuracy (94%) as compared to the other models and the Decision Tree had recorded the lowest accuracy (90%). As shown in Figure 10, the accuracy of our FL-based DL models was even

TABLE 11 | Performance of proposed collision avoidance model on dataset (Onesimu et al. 2021).

Algorithm	Precision	Recall	F1-score	Accuracy (%)
CNN + FedAvg	0.96	1.00	0.98	96.5
LSTM + FedSGD	0.97	0.84	0.95	97.2
DRL + FedPPO	0.98	0.93	0.94	98.0
MLP + FedLU	0.98	0.94	0.91	97.8
GNN + FedGC	0.99	0.97	0.98	98.5

TABLE 12 | Performance of proposed collision avoidance model on generated dataset.

Algorithm	Precision	Recall	F1-score	Accuracy (%)
CNN + FedAvg	0.97	1.00	0.98	97.0
LSTM + FedSGD	0.97	0.92	0.98	97.65
DRL + FedPPO	0.98	0.96	0.94	98.3
MLP + FedLU	0.99	0.96	0.94	98.5
GNN + FedGC	0.99	0.98	0.99	98.8

higher compared to that of the Traffic Accident Prediction System (Adnan, Umer, et al. 2025) when run on the same dataset (Onesimu et al. 2021). The comparison of various DL algorithms in our proposed model, as illustrated in Figure 11, reveals that the proposed models are more accurate in accident predictions, utilising heterogeneity-based vehicle data and RSS and outperform earlier state-of-the-art models.

Figure 11 illustrates the accuracy of the proposed DL models on the simulated dataset, which considers HB parameters and the most accurate parameter types in dataset (Onesimu et al. 2021) in terms of vehicular heterogeneity and RSS. This analysis reveals that incorporating HBF significantly enhances the system's performance and real-life driving conditions play a crucial role in predicting accidents. The accuracy of DL algorithms implemented on both dataset (Onesimu et al. 2021) and the generated dataset in our proposed CAS is shown in the comparison chart in Figure 12. It also demonstrates that optimising the appropriate parameters and generalising features in the data significantly improves prediction accuracy. HBF also play a significant role in prediction accuracy, making it valuable to incorporate these factors to enhance the effectiveness of collision avoidance. Figure 13 shows the performance evaluation of our models with existing models (Khan and Ahmed 2023; Tao et al. 2024).

The effectiveness of the achieved findings demonstrates the feasibility of the proposed FL-based DL models in the real world. It led to the enhanced use of IoV systems, as well as a step towards more sophisticated use of intelligent transportation technologies.

6 | Discussion

In the Internet of Vehicles (IoV), CASs are crucial for improving road safety, preventing accidents and increasing efficiency. The main challenges posed by vehicle differences, environmental factors and HB are addressed by our proposed Federated Learning (FL)-based Deep Learning (DL) framework for

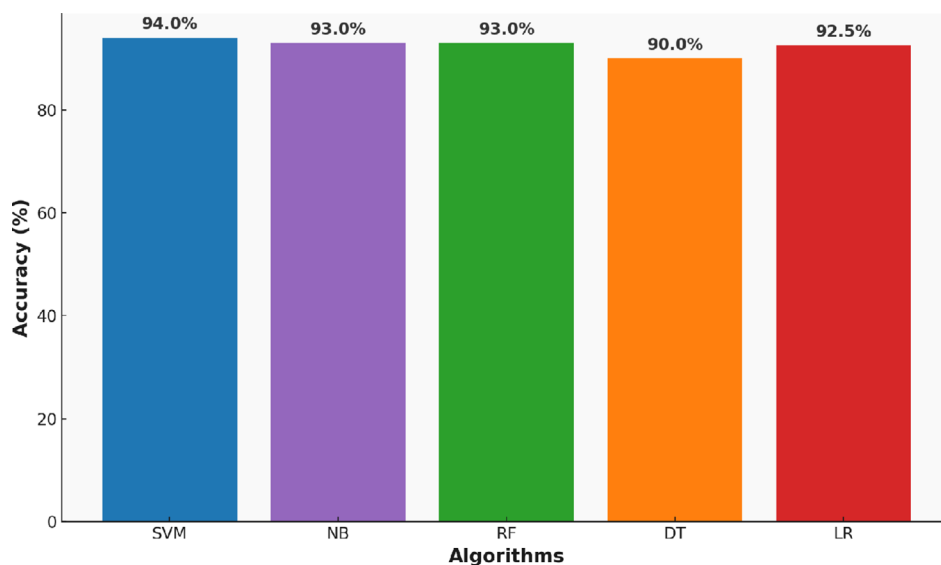


FIGURE 9 | Accuracies of different machine learning models on the dataset (Onesimu et al. 2021).

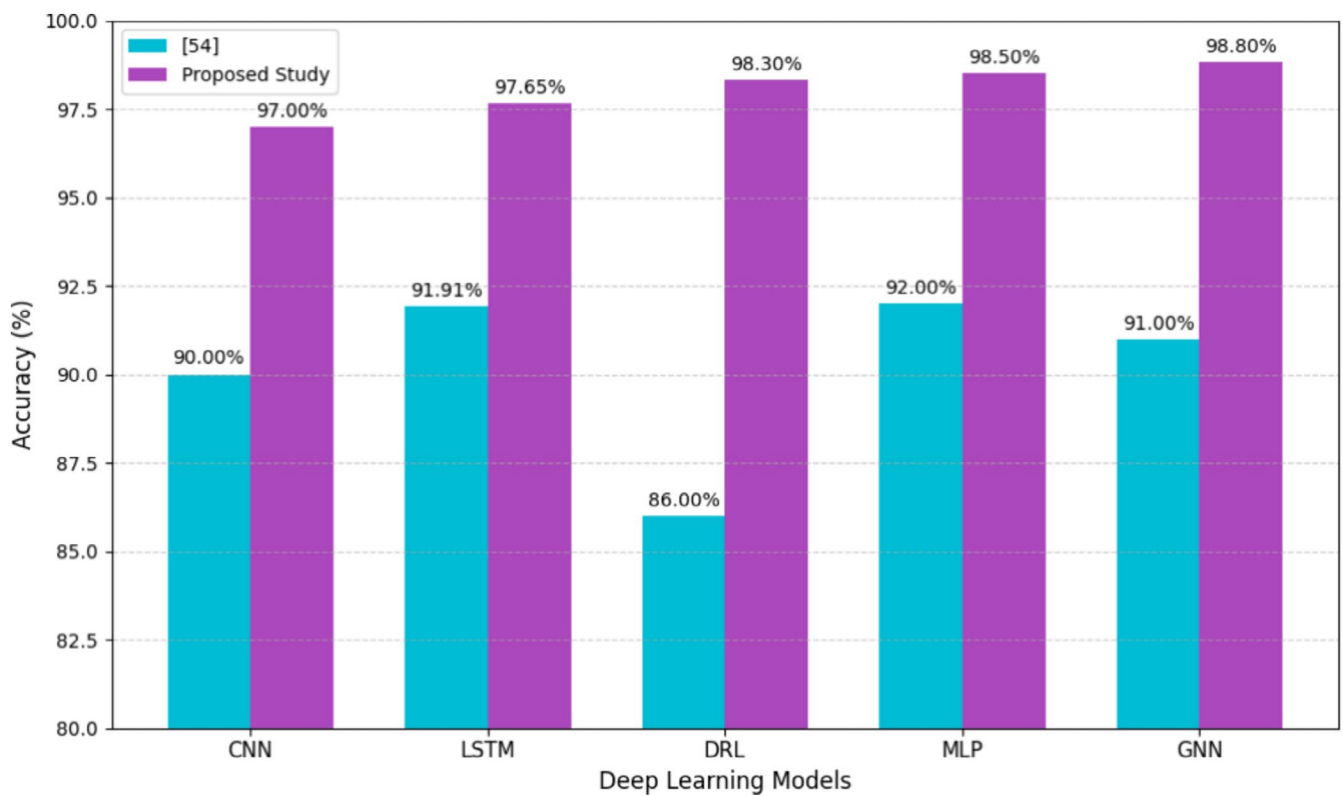


FIGURE 10 | Performance evaluation of our models with existing models w.r.t the parameters from dataset (Onesimu et al. 2021).

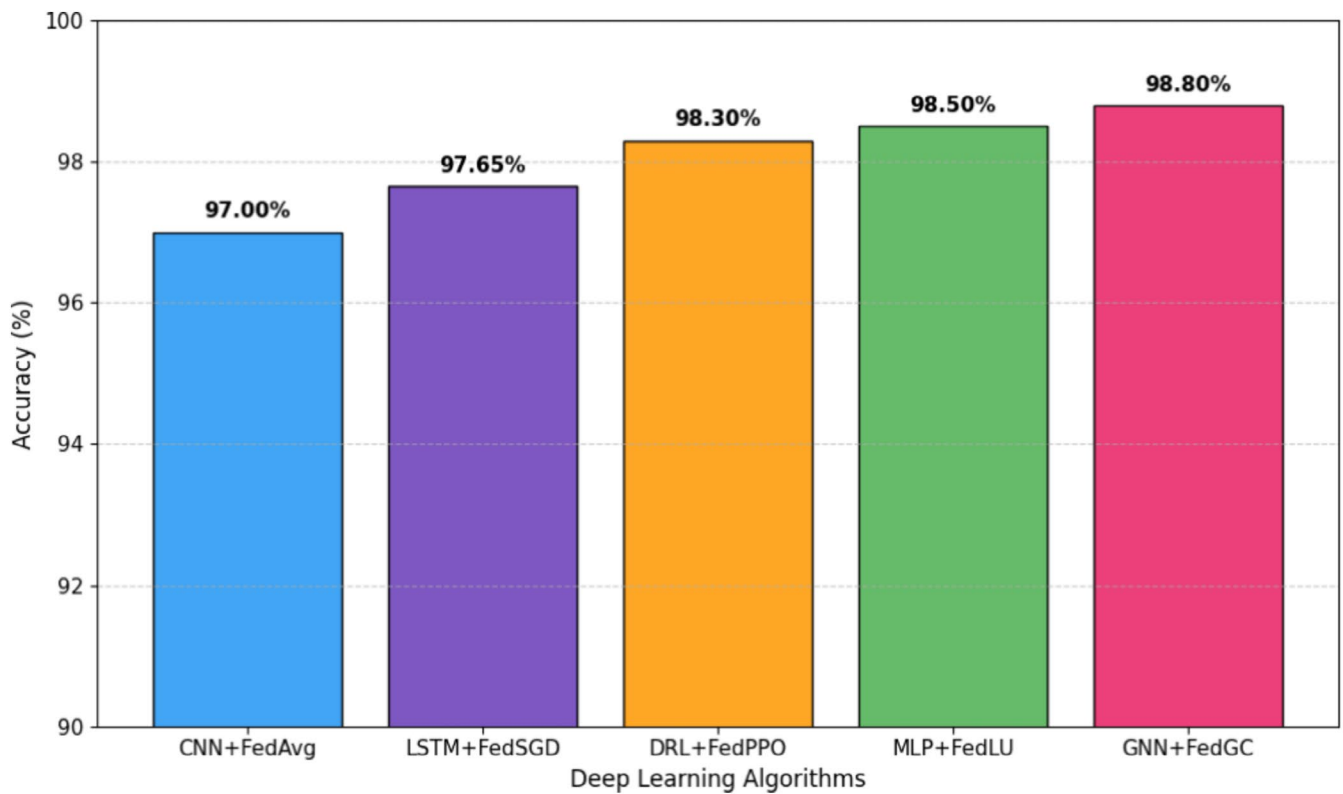


FIGURE 11 | Accuracies of deep learning models on generated dataset.

collision avoidance. This system helps reduce the severity of collisions and prevent accidents, ultimately protecting lives and making transportation safer. By utilising Federated Learning,

our model also facilitates seamless vehicle communication, ensures privacy protection and promotes scalability, thereby contributing to the advancement of intelligent vehicular networks

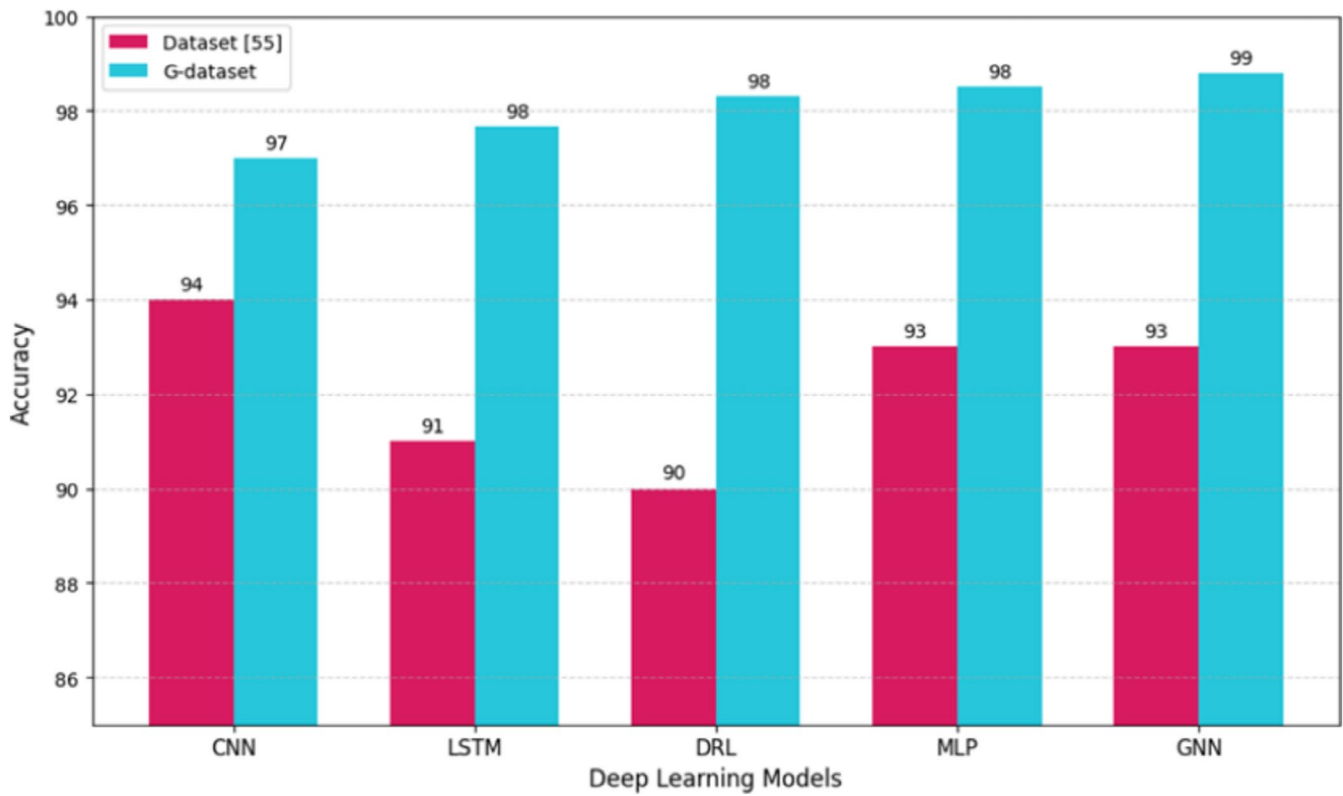


FIGURE 12 | Comparison of deep learning models applied on dataset (Onesimu et al. 2021) and generated dataset.

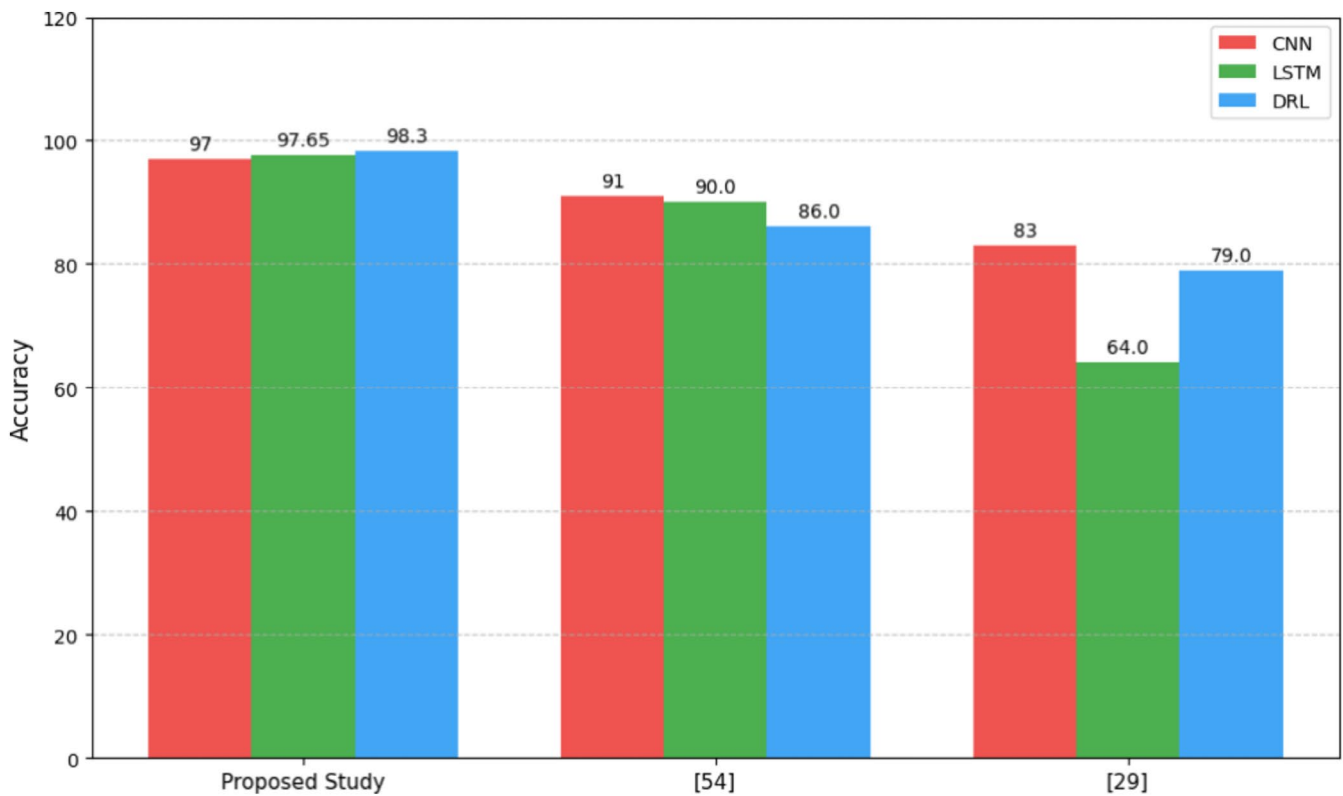


FIGURE 13 | Performance evaluation of our models with existing models.

and the broader development of IoV systems. Digital Twins have definitely taken our collision avoidance model to the next level. However, as IoV systems scale up with more vehicles on the

road, we'll need to continue improving the real-time simulation capabilities of Digital Twins to account for even more variables. For instance, larger urban environments with diverse vehicle

types could push the limits of how many digital twins can be simulated at once. While these simulations are powerful, there's still room for improvement in efficiency. Future work will explore ways to scale up and enhance the realism of these simulations, ensuring they can effectively handle the complexities of a rapidly growing IoV network.

While our results show promising outcomes in predicting and avoiding collisions, it's important to note some limitations that might affect the scalability and generalisation of the findings. One limitation is the availability and variability of real-world IoV data, which can restrict how well the Federated Learning approach scales. Although we consider key factors such as vehicle heterogeneity, weather conditions and road surface types, gathering a broad and representative dataset from various driving environments remains a significant challenge. This issue of data diversity is also highlighted by studies like (Adnan, Umer, et al. 2025), which pointed out that insufficient or biased datasets can limit how well a model generalises to different traffic conditions. Additionally, the model's performance might vary depending on traffic environments, geographic locations or dynamic environmental changes. Another challenge is the rapid evolution of IoV technologies and communication protocols. As these technologies advance, new, unforeseen challenges such as the development of sophisticated vehicular technologies or changes in regulations may affect the model's performance. Ullah et al. (2025) also discussed how emerging communication protocols might impact the effectiveness of existing models in IoV systems. Despite these challenges, understanding these limitations is crucial to improving the robustness and effectiveness of CASSs. Our study also highlights the importance of HB, including driver fatigue, distractions and the diversity of vehicles, which significantly contribute to predicting accidents. These factors are often overlooked in traditional CASSs, which mainly focus on road conditions and vehicle speed. Our approach, which includes HB modelling, sets us apart from earlier studies like (Qin et al. 2025), which did not account for HB in their predictions. By integrating these factors, our system can enhance road safety, ensure regulatory compliance and potentially reduce accidents on the road.

7 | Conclusion

In this research study, we introduced an FL-enabled DL framework for collision avoidance in the IoV, incorporating vehicular diversity, driver behaviour and environmental conditions. The proposed framework utilised multiple DL models, including CNN, LSTM, DRL, MLP and GNN, within an FL context, preserving data privacy, reducing transmission overhead and supporting real-time scalability. A key feature was the inclusion of digital twins, enabling real-time simulations of vehicle behaviour and environmental conditions, improving prediction accuracy without compromising privacy. The experimental results validated the performance of our model, with CNN + FedAvg achieving 97.0%, LSTM + FedSGD 97.65%, DRL + FedPPO 98.3%, MLP + FedLU 98.5% and GNN + FedGC 98.8% precision, surpassing conventional methods in prediction precision and deployment potential. The system incorporated behavioural indicators, such as driver fatigue, yawning, and

alcohol influence, and linked them to vehicle and road parameters for more accurate predictions. FL ensured that sensitive data remained decentralised, making the solution scalable and privacy-preserving. Additionally, the high accuracies achieved by models like GNN + FedGC and DRL + FedPPO demonstrate the effectiveness of temporal and relational learning in dynamic traffic environments. Our system supports real-time interventions, such as automated braking and speed regulation, which are suitable for smart city integration. Future work will integrate collision prediction with ADAS to enhance driver assistance. AI, ML, IoV connectivity and blockchain will be explored to build a smarter, safer, and more trustworthy transportation ecosystem.

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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